The Labor Market Effects of Carbon Pricing

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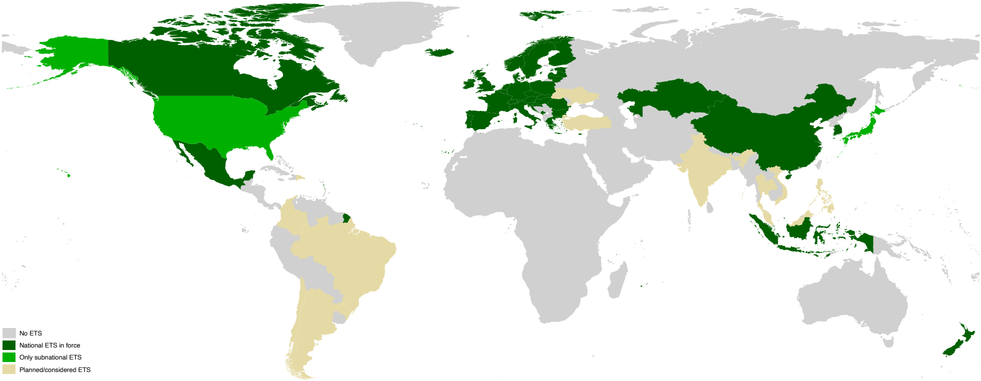
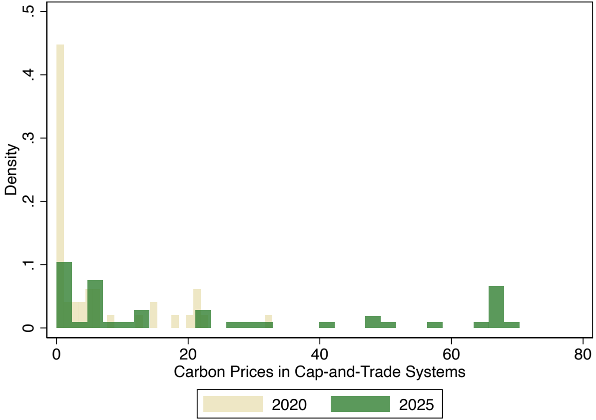
Cap-and-Trade Systems around the World



Cap-and-trade systems are a key carbon mitigation policy instrument

Carbon Prices in Cap-and-Trade Systems



Carbon prices are rising (though not enough to reach net-zero goals)

Carbon Pricing and the Economy

- Carbon prices are likely to further increase in the future
- Debate about the consequences of such increases on the economy overall and on the labor market in particular (UN, ILO, EU)

$\Rightarrow$ Overall employment effects? Winners and losers?

$\rightarrow$ The design of the market

$\rightarrow$ Worker skills

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 → The design of the market
 → Worker skills

Our paper: We study how an increase in **EU ETS carbon prices** affects **workers**

- Rich population-level data sets from the Netherlands
 → Employee-employer matched data, worker, and plant-level characteristics
- Exploit the 2017 reform in the EU ETS that reduced the supply of emission permits
- Design: Matched difference-in-differences

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Document the importance of the design of the cap-and-trade system and worker skills for labor market implications of carbon pricing

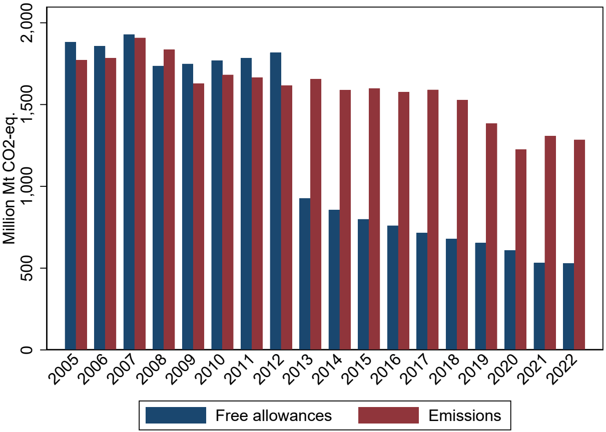
European Union Emissions Trading System

The EU ETS is a cap-and-trade program

- The EU sets an annual emission amount and issues allowances accordingly
- Allowances are distributed either via auctions or free allocations
- Participation is at the installation (not firm) level
- Based on capacity thresholds: e.g., combustion >20 MW total rated thermal input; 2.5t of production per hour for iron and steel manufacturing, etc.
- Firms submit their allowances by April 30 for the previous year
  $\rightarrow$ Participants can keep or sell their unused permits
  $\rightarrow$ Not submitting leads to a fine of 100 euros per tonne + allowance
- Until 2013, the vast majority of allowances were granted for free to installations meeting certain criteria

European Union Emissions Trading System (Cont'd)

- Phase 3: Single, EU-wide cap on emissions replaced previous system of national caps
- Sharp reduction in free permits



The 2017 Reform

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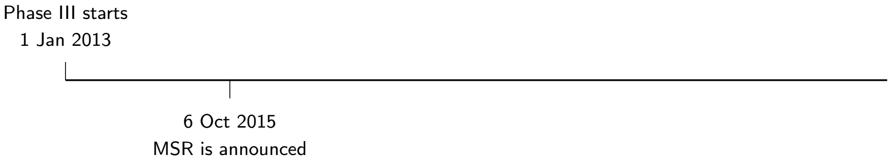

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  - Absorption of 24% of unused allowances instead of 12% if unused is above a threshold
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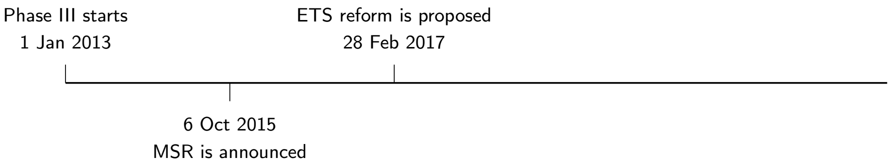

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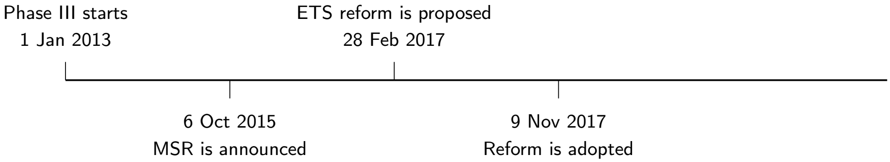

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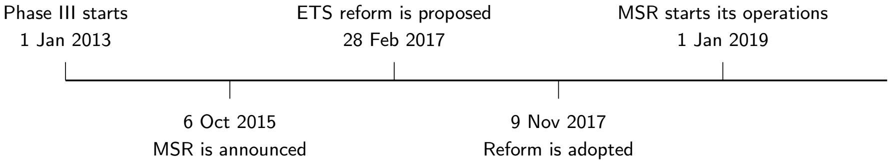
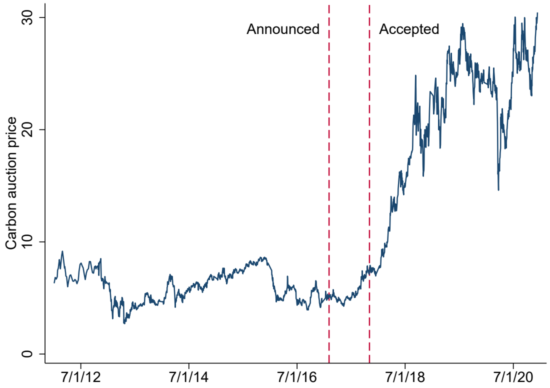


Carbon Prices

- Stark increase in carbon prices following 2017 announcement (De Jonghe et al. 2020, Apicella & Fabiani 2023, Boeckx et al. 2025)



- In our sample, 154.6 permits per worker
- A price increase ≈20 euros → 3 thousand euros shock

Data

- ETS, labor market, firm characteristics, individual characteristics
    1. ETS transactions log: Carbon emissions, free allowances (EUTL)
    2. Labor market: Wages, hours obtained from employee-employer matched data, employment status (CBS)
    3. Firm characteristics: Balance sheet, income statement, sector (CBS)
    4. Individual characteristics: Education, age, gender (CBS)
    5. Plant-level production statistics (CBS)

Empirical Strategy – Matching

- 212 ETS installations/plants belonging to 181 firms
- ETS firms are larger and more profitable, workers are older and earn more

⇒ Use a matching approach to make workers in ETS and non-ETS firms more comparable

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Matching is done at the worker and firm levels. Specifically:

- Start with workers aged 25 to 60 and employed at same company in 2015 and 2016
- Match exactly on industry, gender, high-tenure dummy, all in 2016
- Within each cell, estimate propensity score of being at ETS firm, as function of: age_{t-1}, $\log(\text{hr. wage}_{t-1})$, $\log(\text{hr. wage}_{t-2})$, $\log(\text{firm size})_{t-1}$, and firm $\text{profit/employee}_{t-1}$
- For every treated worker, select non-ETS worker with closest propensity score, with caliper 0.025

Balance Test

- Matching removes the differences between the ETS and non-ETS groups

	Full Sample			*Matched Sample*		
Variable	**Control**	**Treated**	**Diff**	**Control**	**Treated**	**Diff**
Age_{t-1}	40.261	41.196	0.934	42.476	42.495	0.019
	(0.105)	(0.317)	(0.333)	(0.353)	(0.268)	(0.443)
$\log(\text{Hourly Wage}_{t-1})$	3.019	3.354	0.335	3.358	3.371	0.013
	(0.008)	(0.016)	(0.018)	(0.046)	(0.030)	(0.055)
$\log(\text{Hourly Wage}_{t-2})$	2.965	3.293	0.328	3.310	3.323	0.013
	(0.009)	(0.015)	(0.018)	(0.046)	(0.029)	(0.055)
$\log(\text{Size}_{t-1})$	5.494	8.375	2.881	6.426	6.443	0.017
	(0.137)	(0.274)	(0.306)	(0.178)	(0.125)	(0.218)
$\text{Profits}_{t-1}/\text{Workers}_{t-1}$	20.170	45.403	25.233	90.652	86.894	-3.758
	(1.285)	(13.294)	(13.319)	(14.575)	(13.700)	(19.974)
Observations	1,670,567	95,441	1,766,008	17,541	17,541	35,082

Note: all continuous variables are winsorized at 1st and 99th percentile throughout.

Empirical Strategy

Exploit the increase in carbon prices in a matched difference-in-differences setting:

$$y_{it} = \boldsymbol{\beta}\, ETS_i \times Post_t + \gamma_i + \delta_t + \epsilon_{it},$$

where:

- $ETS_i = 1$ for workers in ETS firms as of 2016, $ETS_i = 0$ for matched units
- $Post_t = 1$ if year ≥ 2018
- y_{it}: log(hourly wage) (but also log(wages), earnings, and employment)
- γ_i and δ_t are worker and year fixed effects, respectively
- Throughout, cluster standard errors at 2016-firm level

Event-study version:

$$y_{it} = \sum_{\tau=-3}^{3} \boldsymbol{\beta_\tau}\, ETS_i \times \mathbb{1}(t = t^* + \tau) + \gamma_i + \delta_t + \epsilon_{it},$$

where $\beta_{-1} = 0$

Baseline Results

- No adverse effects on workers – if anything, workers in ETS firms have better outcomes (in line with Dechezlepretre et al. 2023, Colmer et al. 2025, Gallé et al. 2025)
- Significant yet small positive effect on employment

Dep. Var.:	Log(Hr. Wage)	Earnings	Log(Wage)	Employed
	(1)	(2)	(3)	(4)
ETS $\times$ Post	0.018	2,105.471	0.019	0.006**
	(0.016)	(1,482.941)	(0.017)	(0.003)
Observations	240,047	245,574	240,047	245,574
Worker FE	X	X	X	X
Year FE	X	X	X	X

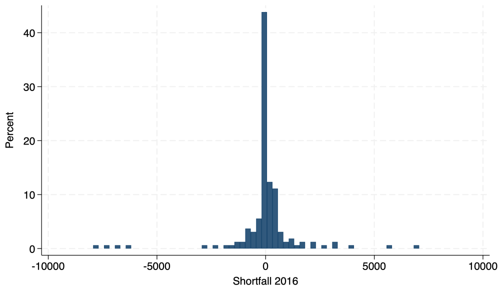
Differences in Permit Shortfall/Surplus

- Firms receive different amounts of free permits, depending on their industry and activity level.
  $\rightarrow$ permit shortfall: emissions-free permit
  $\rightarrow$ Within the same industry, more carbon efficient firms have smaller permit shortfall
- The effect of carbon price likely heterogeneous depending on firm's permit shortfall
- If this difference is positive, the firm systematically uses more permits than it receives; hence, a higher price will penalize it
- We normalize the difference by the number of workers in the plant
  - In the regressions, we demean and standardize it for ease of interpretation

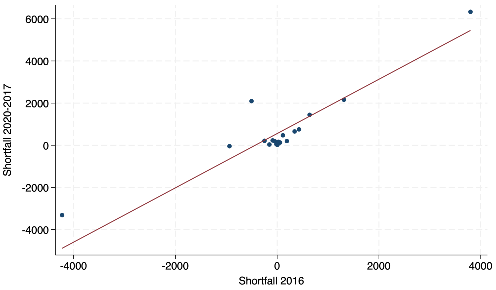
Distribution of Permit Shortfall

- Permit shortfall prior to the shock is a strong predictor of future shortfall as well

A. Distribution of Shortfall in 2016



B. 2020–2017 Shortfall vs 2016 Shortfall



Note: Four plants with shortfall>50,000 are excluded.

Permit Shortfall

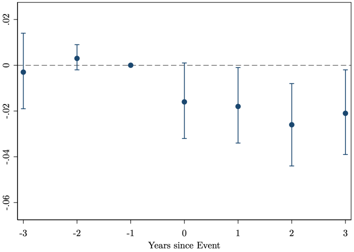
- Workers in ETS firms with permit shortfalls experience a relative reduction in their wages
  $\rightarrow$ 1 std dev higher permit shortfall $\Rightarrow$ 1.8 percent differential decline in hourly wages

Dep. Var.:	Log(Hr. Wage)	Earnings	Log(Wage)	Employed
	(1)	(2)	(3)	(4)
ETS × Post	0.018	2,105.471	0.019	0.006**
	(0.016)	(1,472.398)	(0.017)	(0.003)
Post × Shortfall (Standardized)	0.005	284.042	0.003	0.000
	(0.005)	(480.042)	(0.005)	(0.002)
ETS × Post × Shortfall	-0.018***	-1,372.780**	-0.028***	-0.003
	(0.006)	(560.164)	(0.009)	(0.002)
Observations	240,047	245,574	240,047	245,574
Worker FE	X	X	X	X
Year FE	X	X	X	X

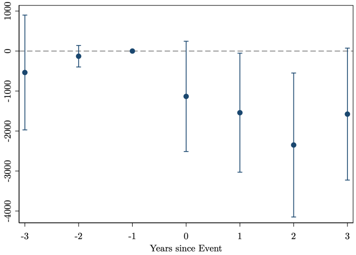
Note: for control group workers, the Shortfall of the matched treatment group worker is used.

Permit Shortfall: Event-Study Evidence

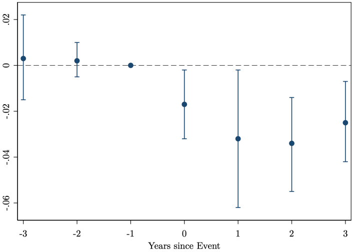
A. log(Hourly Wage)



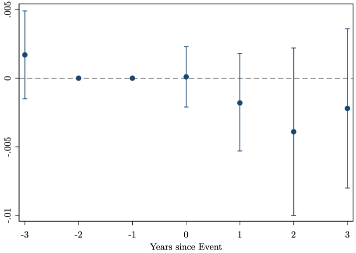
B. Earnings



C. log(Wage)



D. Employed



Heterogeneity in Worker Skills/Education

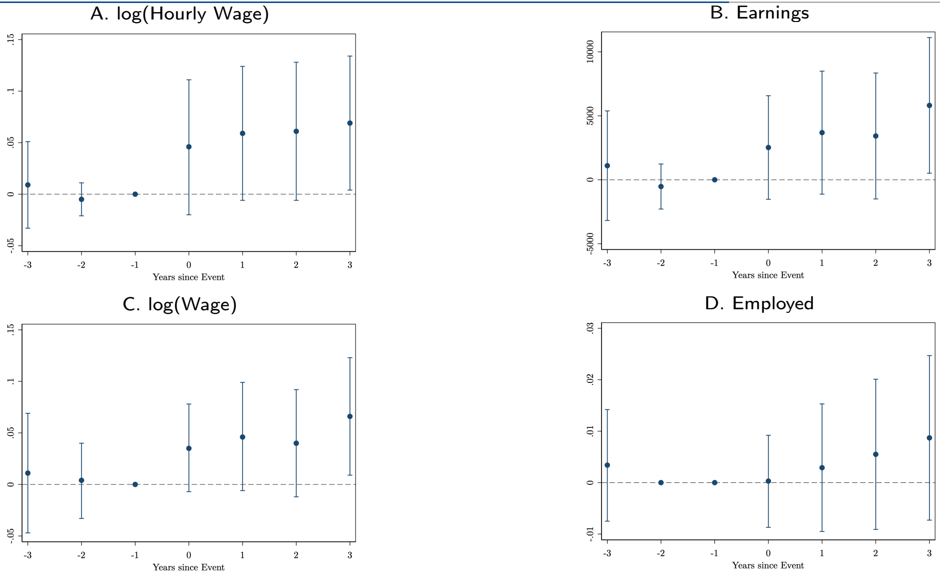
- STEM workers are the most valuable to cut emissions (Vona et al. 2018, Saussay et al. 2022)
- The carbon price shock may increase the value of their emission reduction skills, increasing their value for their employers

STEM and Non-STEM Workers

- In line with idea that their skills become more valuable, STEM workers in ETS firms enjoy a wage increase
  $\rightarrow$ 6.9 percent higher hourly wages compared to STEM workers in non-ETS firms three years after the event (DDD difference = 5 percent)

Dep. Var.:	Log(Hr. Wage)		Earnings		Log(Wage)		Employed	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ETS $\times$ Post	0.016	0.050**	1,981.220	3,533.814*	0.018	0.038	0.006**	0.005
	(0.017)	(0.024)	(1,492.998)	1,821.845	(0.017)	(0.024)	(0.003)	(0.006)
Observations	221,913	18,134	227,003	18,571	221,913	18,134	227,003	18,571
$\beta_{STEM} - \beta_{NO\ STEM}$	0.035		1,552.594		0.021		-0.001	
	(0.022)		(1,449.177)		(0.019)		(0.005)	
$\beta_{STEM} - \beta_{NO\ STEM}$	0.050**		2,420.570*		0.042***		-0.001	
(with Firm-Year FE)	(0.016)		(1,397.933)		(0.016)		(0.006)	
STEM	No	Yes	No	Yes	No	Yes	No	Yes

STEM Workers: Event-Study Evidence



Caliper 0.05 Caliper 0.01 One Lag Matching

STEM Workers & Outside Options

- When carbon prices rise, transition-relevant skills generate a higher revenue.
- If these skills are also valued by outside employers, the workers who have better outside options can obtain a larger share of the additional worker-specific surplus
  - For these workers, the threat to seek other employers is more credible
- Proxies for the willingness to move to other employers
- **Ex-ante** – moved prior to carbon price increase:

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	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ETS $\times$ Post	0.026	0.077**	2,339.055	4,923.031***	0.026	0.062**	-0.005	0.010
	(0.021)	(0.035)	(2,577.594)	(1,867.189)	(0.026)	(0.028)	(0.009)	(0.008)
Observations	7,611	10,523	7,763	10,808	7,611	10,523	7,763	10,808
$\beta_{Mover} - \beta_{Non\ Mover}$	0.051		2,583.976		0.036		0.015	
	(0.034)		(2,633.434)		(0.030)		(0.013)	
Mover	No	Yes	No	Yes	No	Yes	No	Yes

STEM Workers and Outside Options

- **Ex-post definition** – Moved after carbon price increase
- "Movers" are the workers who experience at least one job-to-job transition *after* the event

Dep. Var.:	Log(Hr. Wage)		Earnings		Log(Wage)	
	(1)	(2)	(3)	(4)	(5)	(6)
ETS $\times$ Post	0.033	0.089**	542.463	10,176.575***	0.007	0.114***
	(0.023)	(0.036)	(1,850.971)	(4,501.059)	(0.022)	(0.048)
Observations	11,880	6,254	11,921	6,650	11,880	6,250
$\beta_{Mover} - \beta_{Non\ Mover}$	0.057**		9,634.213*		0.107**	
	(0.026)		(4,974.257)		(0.049)	
Mover	No	Yes	No	Yes	No	Yes

STEM Workers and Firm-Level Outcomes

- Are the higher wage premiums for STEM workers "justified"?
- We use detailed plant-level production statistics and emission data for within-ETS and industry-year comparison:

$$\log(y_{jt}) = \beta Post_t \times Frac.\, STEM\ Workers_j + \alpha_j + \delta_{ts} + \varepsilon_{jt},$$

where y is either:

- log(Emissions)
- log(Energy Costs)
- j, t, and s index plants, years, and sectors, respectively
- We also look at the fraction of *non*-STEM college graduates as a placebo test
- Fraction of STEM workers in the data: Average=5.6%; Standard deviation=7.1%

STEM Workers and Emissions

- Larger reduction in emissions in plants with a higher fraction of STEM workers
- No analogous results for other college graduates
- One st. dev. increase in Frac. STEM → 9–10 percent reduction in emissions

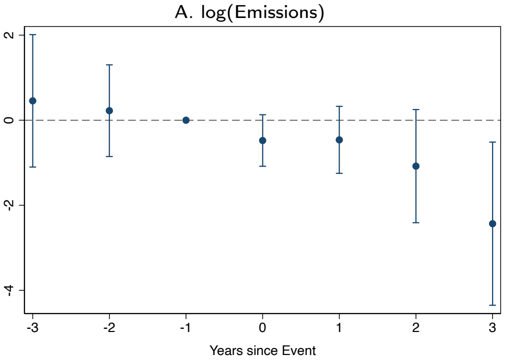
Dep. Var.:	Log(Emissions)					
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Frac. STEM	-1.341**	-1.403**				
	(0.626)	(0.745)				
Post × Frac. Bus.			-0.972	-0.761		
			(0.833)	(0.875)		
Post × Frac. Other					-1.333	-1.129
					(0.961)	(1.031)
Post × Log(Size)		-0.012		-0.022		-0.019
		(0.025)		(0.024)		(0.023)
Post × Profits/Workers		0.000		0.000		0.000
		(0.000)		(0.000)		(0.000)
R^2	0.984	0.984	0.984	0.985	0.985	0.985
Observations	798	798	798	798	798	798
Firm FE	X	X	X	X	X	X
Year-Ind. FE	X	X	X	X	X	X

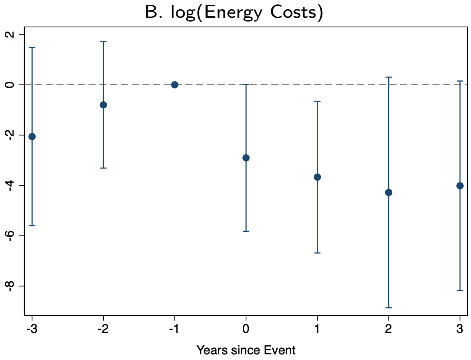
STEM Workers and Energy Costs

- Similar results when looking at energy costs
- One st. dev. increase in Frac. STEM → 12–18 percent reduction in energy costs

Dep. Var.:	Log(Energy Costs)					
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Frac. STEM	-2.566**	-1.725*				
	(1.131)	(0.982)				
Post × Frac. Bus.			-1.671	-0.637		
			(1.204)	(1.111)		
Post × Frac. Other					0.587	1.608
					(1.734)	(1.829)
Post × Log(Size)		-0.086*		-0.098*		-0.117**
		(0.049)		(0.054)		(0.055)
Post × Profits/Workers		0.000		0.000		0.000
		(0.001)		(0.001)		(0.001)
R^2	0.938	0.938	0.937	0.938	0.937	0.938
Observations	794	794	794	794	794	794
Firm FE	X	X	X	X	X	X
Year-Ind. FE	X	X	X	X	X	X

STEM Workers, Emissions, and Energy Costs – Event-Study Evidence





Workforce Composition

- No differential change in firms' STEM worker composition – suggesting rigid labor supply

Dep. Var.:	Fraction STEM	STEM Hire	STEM Separated
	(1)	(2)	(3)
ETS × Post	-0.002	0.008	0.002
	(0.002)	(0.007)	(0.006)
R^2	0.962	0.214	0.133
Obs.	1,288	236,083	247,456
Firm FE	X	X	X
Year FE	X	X	X

Conclusion

- No average losses: The 2017 EU ETS reform caused no overall wage or employment declines
- Distributional effects: Wage impacts vary—permit shortfalls hurt workers; STEM skills yield gains
- Skills matter: STEM-intensive plants reduce energy costs more after the reform
- Rising carbon prices and new ETS systems make skill development and allocation rules central for a fair transition

- Potential future work: complementarity between worker skills and firms' green investments

Thank You!

Free Allocation Formula

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  - Benchmark: Average emission of the best 10 percent installations in that product
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  - Linear reduction: Total allowances decrease every year

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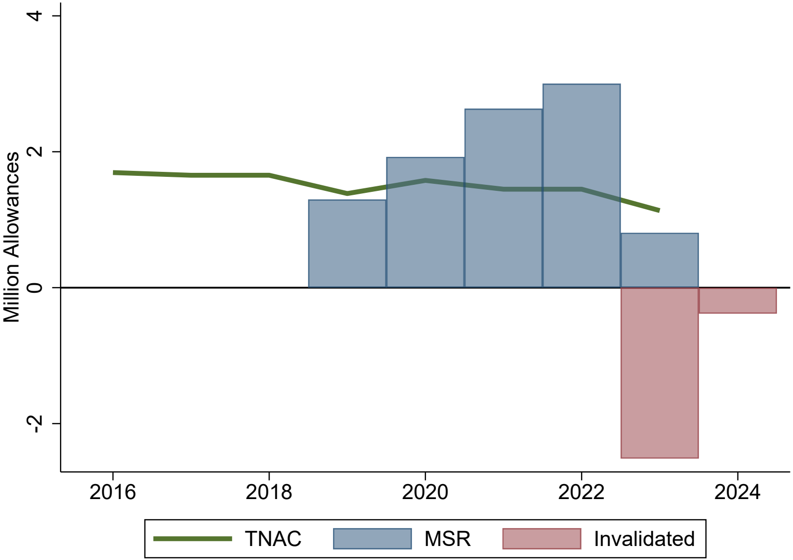
MSR


Million Allowances
4
2
0
-2
2016
2018
2020
2022
2024
TNAC
MSR
Invalidated

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Empirical Strategy – Matching

Top 10 industries with largest number of workers in ETS firms:

Number of Workers by Industry and Size Class Before Matching

Industry	**[1, 49]**	**[50, 249]**	**250+**	**Total**
Air transport	103	133	12,414	12,650
Education	5,646	3,693	60,510	69,849
Electricity, gas, steam and air conditioning supply	488	1,229	9,947	11,664
Human health activities	16,686	6,456	109,358	132,500
Manufacture of basic metals	808	2,004	5,164	7,976
Manufacture of chemicals and chemical products	2,145	5,573	10,912	18,630
Manufacture of food products	4,934	13,365	21,367	39,666
Manufacture of motor vehicles, trailers and semi-trailers	1,107	1,945	6,032	9,084
Manufacture of other non-metallic mineral products	1,733	2,184	2,200	6,117
Manufacture of paper and paper products	804	2,829	1,875	5,508
Other	408,763	333,511	710,090	1,452,364
Total	443,217	372,922	949,869	1,766,008

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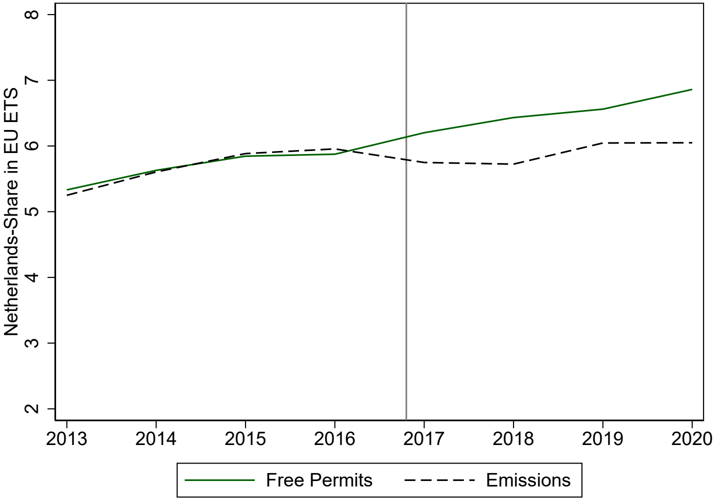
Share of Free Permits and Emissions of Dutch Firms



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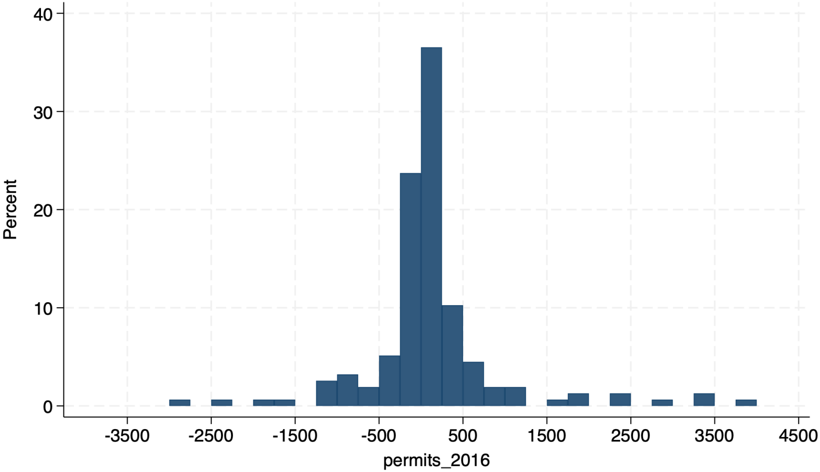
Permit Shortfall – Distribution in 2016



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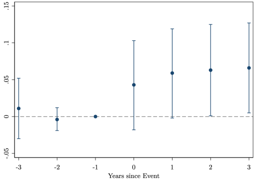
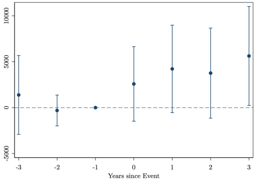
Matching with a 0.01 Caliper

Only STEM Workers

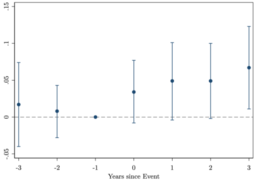
A. log(Hourly Wage)



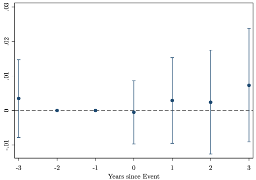
B. Earnings



C. log(Wage)

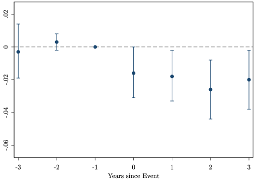


D. Employed

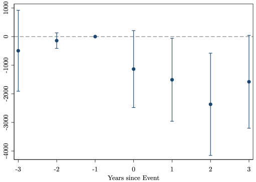


Interactions with Permit Shortfall

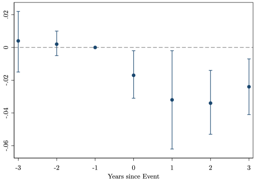
A. log(Hourly Wage)



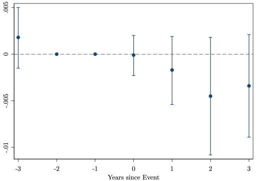
B. Earnings



C. log(Wage)



D. Employed



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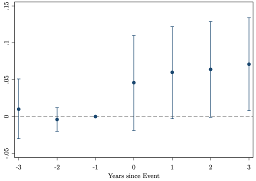
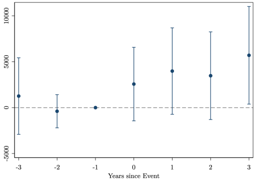
Matching with a 0.05 Caliper

Only STEM Workers

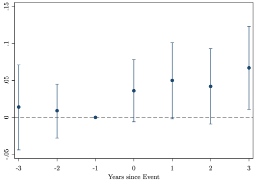
A. log(Hourly Wage)



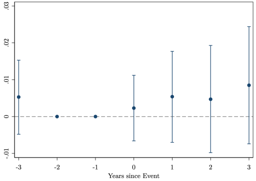
B. Earnings



C. log(Wage)

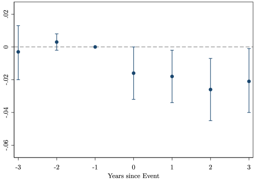


D. Employed

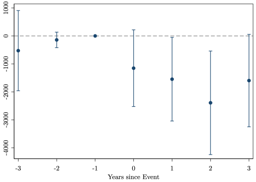


Interactions with Permit Shortfall

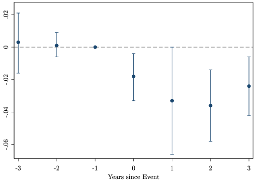
A. log(Hourly Wage)



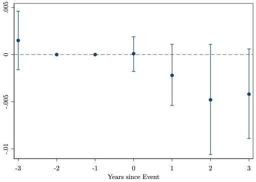
B. Earnings



C. log(Wage)



D. Employed



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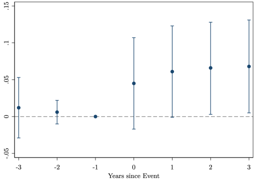
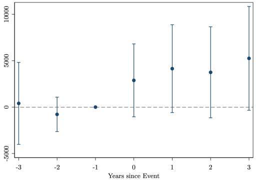
Matching on one Lag of log(Hourly Wage)

Only STEM Workers

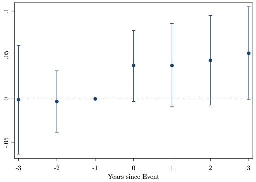
A. log(Hourly Wage)



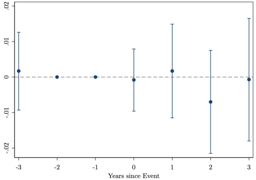
B. Earnings



C. log(Wage)

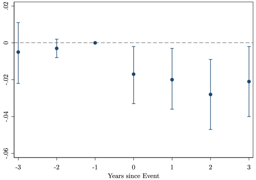


D. Employed

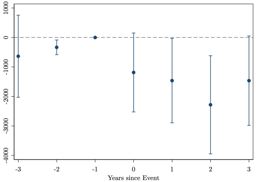


Interactions with Permit Shortfall

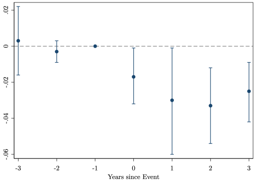
A. log(Hourly Wage)



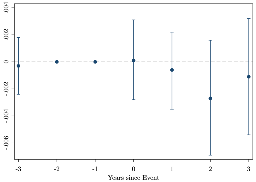
B. Earnings



C. log(Wage)



D. Employed



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